

Quadratic reciprocity

Let $a, p \in \mathbb{Z}^{>0}$, p odd prime. Let $\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a square mod } p \\ -1 & \text{o/w} \end{cases}$

be the Legendre symbol. Then by Euler's criterion, $\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$.

Note that in $\mathbb{Z}/p\mathbb{Z}^\times$, there are exactly $\frac{p-1}{2}$ quadratic residues. By Fermat's little theorem, $a^{p-1} \equiv 1 \pmod{p}$, i.e. $(a^{\frac{p-1}{2}} - 1)(a^{\frac{p-1}{2}} + 1) \equiv 0 \pmod{p}$.

(Clearly, quadratic residues make the first factor vanish, hence they are all its roots. Therefore, the quadratic non-residues make the second factor vanish. $\square$)

Let's now calculate the product Π of all elements in the group $R = \mathbb{Z}/p\mathbb{Z}^\times \times \mathbb{Z}/q\mathbb{Z}^\times / \pm 1$:

We have $R = \{(i, j) : i = 1, \dots, p-1; j = 1, \dots, q-1\}$. Hence,

$$\Pi = \left((p-1)!^{\frac{q-1}{2}}, \left(\frac{q-1}{2}\right)!^{p-1} \right) = \left((p-1)!^{\frac{q-1}{2}}, (q-1)!^{\frac{p-1}{2}} (-1)^{\frac{p-1}{2} \frac{q-1}{2}} \right),$$

since $(\frac{q-1}{2})!^2 = (\frac{q-1}{2})! (\frac{q-1}{2})! \equiv (-1)^{\frac{q-1}{2}} (\frac{q-1}{2})! (q - \frac{q-1}{2})! \equiv (-1)^{\frac{q-1}{2}} (q-1)!$. "upwards factorial to q"

On the other hand, since $R \cong \mathbb{Z}/pq\mathbb{Z}^\times / \pm 1$, we have

$$R = \{(k, k) : k = 1, \dots, \frac{pq-1}{2}; (k, pq) = 1\},$$

so the first factor of Π is

$$\frac{\left(\prod_{i=1}^{p-1} i\right) \left(\prod_{p+i}^{p-1} p+i\right) \dots \left(\prod_{\frac{q-1}{2}-1}^{\frac{q-1}{2}-1} p+i\right) \left(\prod_{\frac{q-1}{2}}^{\frac{q-1}{2}} p+i\right)}{1 \cdot q \cdot 2q \cdot \dots \cdot \frac{p-1}{2} q} \stackrel{\left(\frac{q}{p}\right)}{\equiv} \frac{(p-1)!^{\frac{q-1}{2}}}{q^{\frac{p-1}{2}}},$$

and similarly for the second factor. Hence,

$$\Pi = \left((p-1)!^{\frac{q-1}{2}} \left(\frac{q}{p}\right), (q-1)!^{\frac{p-1}{2}} \left(\frac{p}{q}\right) \right).$$

Comparing to the first form of Π , we get recall the definition of R

$$\left(1, (-1)^{\frac{p-1}{2}} (-1)^{\frac{q-1}{2}}\right) = \pm \left(\left(\frac{q}{p}\right), \left(\frac{p}{q}\right)\right).$$

Hence, we get quadratic reciprocity $\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \frac{q-1}{2}}$.

Let $s_0 = 4$, $s_{i+1} = s_i^2 - 2$. Let $q = 2^p - 1$. ↗ 2 Then q is prime iff $s_{p-2} \equiv 0 \pmod{q}$.

Let $w = 2 + \sqrt{3}$, $\bar{w} = 2 - \sqrt{3}$. Note first that $s_i = w^{2^i} + \bar{w}^{2^i}$. Now:

⇐ Assume $s_{p-2} \equiv 0 \pmod{q}$, i.e. $w^{2^{p-2}} + \bar{w}^{2^{p-2}} = kq$. Then

$$(*) \quad w^{2^{p-1}} = kq w^{2^{p-2}} - 1. \quad \leftarrow (w\bar{w})^{2^{p-2}}$$

Since q is odd, any prime divisor $r|q$ is > 2 . Let $R = \mathbb{Z}/r\mathbb{Z}[\sqrt{3}]$. Then $|R| = r^2$ and $|R^\times| \leq r^2 - 1$. Also, $kq w^{2^{p-2}} = 0$ in R , so

$$w^{2^{p-1}} = -1,$$

by (*). Hence, $w \in R^\times$ and $\text{ord}(w) \nmid 2^{p-1}$ but $w^{2^p} = 1$. Therefore,

$$2^p \leq |R^\times| \leq r^2 - 1 < r^2 \leq q = 2^p - 1, \quad \leftarrow \text{if } q \text{ is composite}$$

which is a contradiction.

⇒ We have $2^p - 1 \equiv 1 \pmod{3}$, $2^p - 1 \equiv 3 \pmod{4}$.

By quadr. reciprocity, $\left(\frac{3}{q}\right) \left(\frac{q}{3}\right) = (-1)^{\frac{3-1}{2} \frac{q-1}{2}} = -1$,

and since $q \equiv 1 \pmod{3}$ is a quadr. residue, $\left(\frac{3}{q}\right) = -1$. By Euler's criterion, $3^{\frac{q-1}{2}} \equiv -1 \pmod{q}$.

Since $2^p \equiv 1 \pmod{q}$, we get $2 \equiv 2 \cdot 2^p = (2^{\frac{p+1}{2}})^2 \pmod{q}$ and hence, by Euler's criterion, $2^{\frac{q-1}{2}} \equiv 1 \pmod{q}$. Hence,

$$24^{\frac{q-1}{2}} = (2^{\frac{q-1}{2}})^3 \cdot 3^{\frac{q-1}{2}} \equiv -1 \pmod{q}.$$

Let $R = \mathbb{Z}/q\mathbb{Z}[\sqrt{3}] \ni \sigma = 2\sqrt{3}$. Then in R ,

$$\begin{aligned} (6 + \sigma)^q &= 6^q + \sigma^q \\ &= 6 + 2\sqrt{3}^q = 6 + 2 \cdot 3^{\frac{q-1}{2}} \sqrt{3} \\ &= 6 - \sigma. \end{aligned}$$

Since $w = \frac{1}{24} (6 + \sigma)^2$, we have

$$w^{\frac{q+1}{2}} = \frac{(6 + \sigma)(6 + \sigma)^q}{24 \cdot 24^{\frac{q-1}{2}}} = \frac{(6 + \sigma)(6 - \sigma)}{-24} = -1.$$

Therefore, $w^{\frac{q+1}{2}} \bar{w}^{\frac{q+1}{2}} = -\bar{w}^{\frac{q+1}{2}}$,

i.e. $w^{\frac{q+1}{2}} + \bar{w}^{\frac{q+1}{2}} = 0$. Substituting $q = 2^p - 1$, we get

$$s_{p-2} = w^{2^{p-2}} + \bar{w}^{2^{p-2}} = 0$$

in R , and hence in $\mathbb{Z}/q\mathbb{Z}$. $\square$